ECO 310: Precept 5

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Problem 1: Market of Infinitesimally Small Firms

Consider a market of infinitesimally small firms, where the firm with index $a \geq 0$ has cost function

$$C_a(q) = \frac{a^2}{2} + \frac{q^2}{2}.$$

To compute market supply, use the fact that for every $m > 0$, there is a mass m of firms with indices in $[0, m]$. (For practical purposes, just take this to mean that there are m firms with indices in $[0, m]$ and don't worry about the possibility of a non-integer quantity). Firms are price takers and face demand curve

$$Q^D(p) = 10 - 3p.$$

(a) Find the short-run shutdown prices for a firm with index a, assume that fixed costs are unavoidable.

(b) Find the SR individual supply curve.

Problem 1(a): Key Concepts

- **Shutdown Rule:** A firm shuts down if $p < \min AVC$.
- **Fixed vs. Variable Costs:**
  - Fixed cost: $\frac{a^2}{2}$ (does not depend on q).
  - Variable cost: $\frac{q^2}{2}$.
- **Average Variable Cost (AVC):**

$$AVC(q) = \frac{\frac{q^2}{2}}{q} = \frac{q}{2}.$$

- **Marginal Cost (MC):** Derivative of the variable cost.
- **Profit Maximization:** A firm produces where $p = MC$, provided $p \geq$ shutdown price.

Problem 1(a): Solution

1. Identify that $\frac{a^2}{2}$ is a fixed cost and $\frac{q^2}{2}$ is the variable cost.
2. Compute the average variable cost:

$$AVC(q) = \frac{q}{2}.$$

3. Note that $AVC(q)$ reaches its minimum at $q = 0$, which is 0.
4. **Answer:** The short-run shutdown price is zero.

   (We did the long-run version in class, finding shutdown price $p = \min AC = a$, achieved by producing $q = a$: this is how to set $MC = AC$, which we learned was the recipe for minimizing AC.)

Problem 1(b): Solution

1. Calculate the marginal cost:
$$MC = \frac{d}{dq}\left(\frac{q^2}{2}\right) = q.$$
2. Apply the first order condition (FOC): set $p = MC$, so
$$p = q,$$
provided $p \geq 0$.
3. **Answer:** The SR supply curve is
$$q(p) = p \quad \text{for } p \geq 0.$$

Problem 2: Question

Calculate the long-run cost function for a firm that produces according to production technology

$$f(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$$

where K is capital and L is labor.

Problem 2: Efficient Input Spending

1. Compute the marginal products:

$$MP_K = \frac{1}{2}K^{-\frac{1}{2}}L^{\frac{1}{2}}, \quad MP_L = \frac{1}{2}K^{\frac{1}{2}}L^{-\frac{1}{2}}.$$

2. Set the bang-per-buck condition:

$$\frac{MP_K}{MP_L} = \frac{L}{K} = \frac{r}{w} \quad \Longrightarrow \quad L = \frac{r}{w}K \quad \text{or} \quad wL = rK.$$

Problem 2: Production Constraint and Input Choices

1. Use the production constraint:
$$K^{\frac{1}{2}} L^{\frac{1}{2}} = q.$$
2. Substitute $L = \frac{r}{w} K$:
$$K^{\frac{1}{2}} \left(\frac{r}{w} K \right)^{\frac{1}{2}} = K \sqrt{\frac{r}{w}} = q.$$
3. Solve for K:
$$K = q \sqrt{\frac{w}{r}},$$
and then for L:
$$L = \frac{r}{w} K = q \sqrt{\frac{r}{w}}.$$

Problem 2: Minimum Cost Calculation

1. Substitute the optimal input choices into the cost function:

$$C^{LR}(q) = rK + wL = r\left(q\sqrt{\frac{w}{r}}\right) + w\left(q\sqrt{\frac{r}{w}}\right).$$

2. Simplify the expression:

$$C^{LR}(q) = q\sqrt{rw} + q\sqrt{rw} = 2q\sqrt{rw}.$$

3. **Answer:**

$$C^{LR}(q) = 2\sqrt{rw} \cdot q.$$

Problem 3: Question

There are three local Christmas tree farms, each with up to 2 trees for sale. Farm #1 incurs a cost of \$90 per tree it cuts down, Farm #2's cost is \$50 per tree, and Farm #3's cost is \$70 per tree. And there are 12 buyers looking for Christmas trees: odd-numbered students 1,3,5,7,9,11 have valuations equal to 20 times their student number, and even-numbered students have valuation zero.

(a) Find all possible equilibrium prices and quantities.

Problem 3(a): Solution

1. Clearing the market left-to-right gives $q^* = 4$ (the first four pairs trade; the last two do not).
2. For the equilibrium price:
   - To encourage the 4th trade: require $70 \leq p^* \leq 100$.
   - To discourage the 5th trade: require $60 \leq p^* \leq 90$.
3. The binding inequalities are:

$$70 \leq p^* \leq 90.$$

Problem 3(b): Question

There are three local Christmas tree farms, each with up to 2 trees for sale. Farm #1 incurs a cost of \$90 per tree it cuts down, Farm #2's cost is \$50 per tree, and Farm #3's cost is \$70 per tree. And there are 12 buyers looking for Christmas trees: odd-numbered students 1,3,5,7,9,11 have valuations equal to 20 times their student number, and even-numbered students have valuation zero.

(b) Now suppose that each tree purchased generates a \$40 positive externality for the buyer's neighborhood, as the neighbors feel happy when they see trees in windows. Calculate the socially optimal number of Christmas tree sales, i.e. that maximizes social value minus social cost.

Problem 3(b): Solution

1. Adjust buyer valuations by \$40:

social values	260	220	180	140	100
60					

2. Compare with seller costs:

costs	50	50	70	70	90
90					

3. The social value exceeds cost for the first 5 units, but not the 6th.
4. **Answer:** The socially optimal number of trees sold is 5.

Problem 3(c): Question

There are three local Christmas tree farms, each with up to 2 trees for sale. Farm #1 incurs a cost of \$90 per tree it cuts down, Farm #2's cost is \$50 per tree, and Farm #3's cost is \$70 per tree. And there are 12 buyers looking for Christmas trees: odd-numbered students 1,3,5,7,9,11 have valuations equal to 20 times their student number, and even-numbered students have valuation zero.

(c) As in (b), each tree purchased generates a \$40 positive externality for the neighborhood. Accordingly, a neighborhood planner decides to encourage tree sales by offering a subsidy of $\$s$ to anyone who purchases a tree. Find all subsidies $\$s$ that will lead to the socially optimal number of tree sales, as in (b).

Problem 3(c): Solution

1. The new buyer valuations with subsidy become:

$$\text{buyers} \quad 220+s \quad 180+s \quad 140+s \quad 100+s \quad 60+s \quad 20+s$$

2. To have the 5th trade occur, we need:

$$60+s \geq 90 \implies s \geq 30.$$

3. To prevent the 6th trade, we require:

$$20+s \leq 90 \implies s \leq 70.$$

4. **Answer:** The subsidy must satisfy

$$30 \leq s \leq 70.$$